FlowMLLab | Week 1.2

Pressure–Velocity Coupling

SIMPLE, PISO and PIMPLE

Finite-volume equations and the lid-driven cavity

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The central problem. Momentum determines how fluid accelerates under pressure and viscous forces. Continuity constrains the resulting velocity. Pressure–velocity coupling obtains fields that satisfy both requirements on the same grid.

Prerequisites. Continuity and component momentum equations, finite differences, a linear system, and the Week 1 streamfunction–vorticity cavity example. Matrix notation is introduced after the scalar finite-volume equations.

Learning outcomes. Derive a cell pressure-correction equation; interpret predictor and corrected velocities; distinguish steady iteration, inner correction and outer correction; evaluate continuity, momentum defects, accuracy and computational work.

Sections	Topics
1–4	Conservation laws, staggered grid, momentum coefficients
5–8	Pressure correction, conservation example and SIMPLE
9–11	Backward Euler, PISO and a numerical correction example
12–15	PIMPLE, nonlinear coefficient updates and convergence
16–21	Recorded cavity comparisons and coupling study
22–23	Notebook controls, exercises and references

Scope. Two-dimensional, incompressible, laminar flow on a uniform staggered grid. The retained calculations compare $Re = 100, 400$ and 1000. Higher-Re spatial accuracy and transient accuracy require their own checks; convergence of a two-dimensional steady solver does not establish the physical stability of a three-dimensional flow.

1 Incompressible flow: conservation laws

Horizontal velocity $u_d(x_d, y_d, t_d)$, vertical velocity $v_d(x_d, y_d, t_d)$ and pressure $p_d(x_d, y_d, t_d)$ are the unknown fields. Subscript d denotes dimensional quantities. For a Newtonian fluid with constant density ρ and constant kinematic viscosity ν, with no body force, continuity is

$$\frac{\partial u_d}{\partial x_d} + \frac{\partial v_d}{\partial y_d} = 0.$$

Conservation of horizontal and vertical momentum gives

$$\frac{\partial u_d}{\partial t_d} + u_d \frac{\partial u_d}{\partial x_d} + v_d \frac{\partial u_d}{\partial y_d} = -\frac{1}{\rho}\frac{\partial p_d}{\partial x_d} + \nu \left(\frac{\partial^2 u_d}{\partial x_d^2} + \frac{\partial^2 u_d}{\partial y_d^2} \right),$$
$$\frac{\partial v_d}{\partial t_d} + u_d \frac{\partial v_d}{\partial x_d} + v_d \frac{\partial v_d}{\partial y_d} = -\frac{1}{\rho}\frac{\partial p_d}{\partial y_d} + \nu \left(\frac{\partial^2 v_d}{\partial x_d^2} + \frac{\partial^2 v_d}{\partial y_d^2} \right).$$

The left sides contain local and convective acceleration. Pressure gradients and viscous diffusion provide the right-side accelerations. These two equations and continuity form the incompressible Navier–Stokes system.

Scales. Let L be cavity side length and U the lid speed. Define $x = x_d/L$, $y = y_d/L$, $t = t_d U/L$, $u = u_d/U$, $v = v_d/U$, and $p = (p_d - p_{\text{ref}})/(\rho U^2)$. Here p_{ref} is a fixed arbitrary reference pressure. Then $Re = UL/\nu$ and

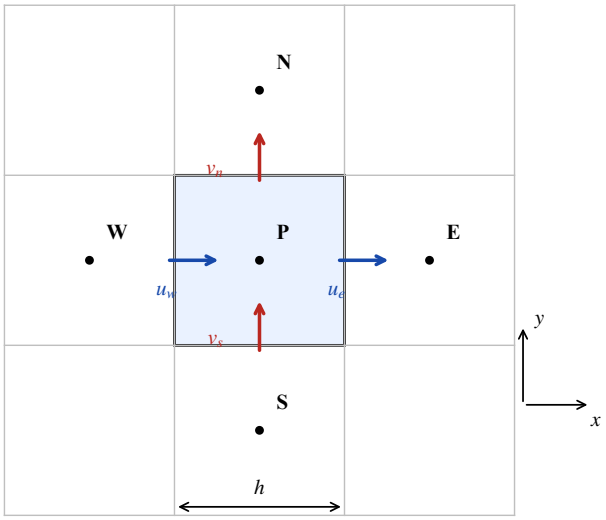
$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0,$$
$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{Re}\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right),$$
$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{Re}\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right).$$

The following sections use these dimensionless quantities. At steady state the local time derivatives vanish, while convection, pressure and diffusion still balance.

Cavity conditions. The unit-square top lid has $u = 1, v = 0$; three stationary walls have $u = v = 0$. Every wall is impermeable. The ideal moving-lid corners contain a velocity discontinuity, so pressure and vorticity extrema there require care. The introductory case is $Re = 100$.

2 Staggered locations and a pressure control volume

The equations in Section 1 are dimensionless. We now integrate them over a small fixed region of the grid. A *control volume* is a chosen region of the flow domain. In a two-dimensional calculation it is an area per unit depth in the third direction.



Pressure p_P is stored at pressure-cell center P. Horizontal velocity is stored at vertical faces; vertical velocity is stored at horizontal faces. For the cell around P, use lowercase e, w, n, s for east, west, north and south faces. Uppercase E, W, N, S denote neighboring *pressure centers*.

First allow a rectangular cell:

$$V_P = [x_w, x_e] \times [y_s, y_n], \qquad \Delta x = x_e - x_w, \quad \Delta y = y_n - y_s, \quad |V_P| = \Delta x \Delta y.$$

Here $|V_P|$ means its area. The boundary is ∂V_P. Its outward unit normal $\boldsymbol{n}$ points away from P at each face:

Face	Outward normal	Face length	Outward velocity component
East	$(1, 0)$	Δy	$+u_e$
West	$(-1, 0)$	Δy	$-u_w$
North	$(0, 1)$	Δx	$+v_n$
South	$(0, -1)$	Δx	$-v_s$

All u values are positive rightward and all v values positive upward. These coordinate conventions do not reverse at the west or south face. The minus signs arise from the *outward normal*, rather than from a different definition of velocity.

The uniform square grid used below has $\Delta x = \Delta y = h = 1/N$, where N is the number of pressure cells along one side. Its cell area is h^2 and each face length per unit depth is h.

2 continued Integral and discrete continuity equations

Start with the differential equation

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

Integrate both terms over the same cell, with $\mathrm{d}V = \mathrm{d}x\,\mathrm{d}y$:

$$\int_{y_s}^{y_n} \int_{x_w}^{x_e} \frac{\partial u}{\partial x}\,\mathrm{d}x\,\mathrm{d}y + \int_{x_w}^{x_e} \int_{y_s}^{y_n} \frac{\partial v}{\partial y}\,\mathrm{d}y\,\mathrm{d}x = 0.$$

For the first term integrate in x while holding y fixed. For the second integrate in y while holding x fixed. The fundamental theorem of calculus gives

$$\int_{y_s}^{y_n} [u(x_e,y) - u(x_w,y)]\,\mathrm{d}y + \int_{x_w}^{x_e} [v(x,y_n) - v(x,y_s)]\,\mathrm{d}x = 0.$$

Thus a derivative inside the cell becomes the difference of values on its two opposite faces. This is also the divergence theorem in two dimensions:

$$\int_{V_P} \nabla \cdot \boldsymbol{u}\,\mathrm{d}V = \int_{\partial V_P} \boldsymbol{u} \cdot \boldsymbol{n}\,\mathrm{d}S = 0.$$

The line element $\mathrm{d}S$ runs along a cell face. Define each face-average velocity by its face integral; for example,

$$\overline{u}_e = \frac{1}{\Delta y} \int_{y_s}^{y_n} u(x_e,y)\,\mathrm{d}y.$$

Replacing a face average by the stored face-center sample is the midpoint approximation used on this grid. With those samples the discrete balance is

$$\underbrace{\Delta y\, u_e}_{\text{east outflow}} - \underbrace{\Delta y\, u_w}_{\text{west inflow}} + \underbrace{\Delta x\, v_n}_{\text{north outflow}} - \underbrace{\Delta x\, v_s}_{\text{south inflow}} = 0.$$

Each term is a signed volume flux per unit depth. Dividing by cell area gives

$$\frac{u_e - u_w}{\Delta x} + \frac{v_n - v_s}{\Delta y} = 0, \qquad \boxed{r_{c,P} = \frac{u_e - u_w + v_n - v_s}{h} = 0 \quad \text{on a square grid.}}$$

The residual $r_{c,P}$ is net outward volume flux divided by area. It is not a pressure change. For constant density ρ, multiplication of every flux by ρ gives the corresponding mass balance.

Numerical check. With $h = 0.25$ and $(u_e, u_w, v_n, v_s) = (0.30, 0.20, 0.10, 0.05)$, net outward flux is $0.25(0.30–0.20+0.10–0.05) = 0.0375$ and $r_{c,P} = 0.0375/0.25^2 = 0.60$. This tentative velocity violates local continuity even if a neighboring cell has an equal and opposite imbalance. Shared internal fluxes cancel in a global sum; they must still balance in each individual cell.

3 Conservative form and integral momentum balance

The horizontal momentum equation from Section 1 is

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{Re}\nabla^2 u.$$

To convert convection into face fluxes, apply the product rule:

$$\begin{aligned}\frac{\partial(u^2)}{\partial x} + \frac{\partial(uv)}{\partial y} &= 2u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + u\frac{\partial v}{\partial y} \\ &= u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + u\underbrace{\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)}_{0 \text{ by continuity}}.\end{aligned}$$

Consequently, for the incompressible continuum equations,

$$\frac{\partial u}{\partial t} + \nabla\cdot(u\boldsymbol{u}) = -\frac{\partial p}{\partial x} + \frac{1}{Re}\nabla\cdot(\nabla u).$$

The scalar u in $u\boldsymbol{u}$ is the horizontal momentum per unit mass being transported; $\boldsymbol{u} = (u, v)$ is the transporting velocity.

Integrate over the *horizontal-velocity* control volume V_e, centered on the stored velocity u_e. It is a different region from pressure cell V_P. For a fixed control volume, the time derivative can pass outside the integral. Apply the divergence theorem separately to convection, pressure and diffusion:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{V_e} u\,\mathrm{d}V + \int_{\partial V_e} u(\boldsymbol{u}\cdot\boldsymbol{n})\,\mathrm{d}S = -\int_{\partial V_e} p\,n_x\,\mathrm{d}S + \frac{1}{Re}\int_{\partial V_e}\nabla u\cdot\boldsymbol{n}\,\mathrm{d}S.$$

Here n_x is the horizontal component of the outward unit normal. The pressure identity used above is

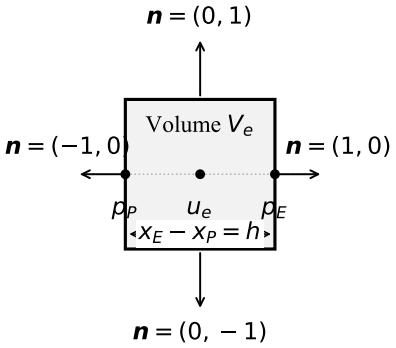
$$\int_{V_e}\frac{\partial p}{\partial x}\,\mathrm{d}V = \int_{\partial V_e} p\,n_x\,\mathrm{d}S.$$

Term	Physical meaning in this equation
Time derivative	Accumulation of horizontal momentum per unit density
Convective integral	Horizontal momentum carried through the boundary by flow
Pressure integral	Horizontal pressure force per unit density
Diffusive integral	Viscous transport of horizontal momentum

These integral balances follow from the differential equations. Face quadrature and interpolation convert them into equations for the stored grid values.

3 continued The momentum volume and its four face contributions

The horizontal velocity u_e lies halfway between pressure centers P and E. Its square momentum volume extends from x_P to x_E, a distance h. The pressure centers therefore lie at the midpoints of its west and east faces.



At this volume's boundaries, write $c_\mathrm{E}, c_\mathrm{W}$ for the horizontal advecting velocities and $c_\mathrm{N}, c_\mathrm{S}$ for the vertical advecting velocities. All are positive in their coordinate directions. Write $\widehat{u}_\mathrm{E}$, $\widehat{u}_\mathrm{W}$, $\widehat{u}_\mathrm{N}$, $\widehat{u}_\mathrm{S}$ for the horizontal component being *transported* at those boundaries. The hat values are interpolated from neighboring velocity samples.

Convection. Midpoint evaluation of the four integrals gives

$$h\left(c_\mathrm{E}\widehat{u}_\mathrm{E} - c_\mathrm{W}\widehat{u}_\mathrm{W} + c_\mathrm{N}\widehat{u}_\mathrm{N} - c_\mathrm{S}\widehat{u}_\mathrm{S}\right).$$

North/south transport is still transport of *horizontal* momentum: the fluid crosses those faces with vertical velocity while carrying horizontal u.

Pressure. Only the east and west faces have $n_x \neq 0$. Approximating their pressure averages by p_E and p_P gives

$$-\int_{\partial V_e} p n_x \,\mathrm{d}S \simeq -h p_E + h p_P = -h(p_E - p_P).$$

After dividing by area h^2, this is $-(p_E - p_P)/h$. The separation between P and E is h, not $2h$.

Diffusion. The normal derivative changes sign on west and south:

$$\frac{h}{Re}\left[\left(\frac{\partial u}{\partial x}\right)_\mathrm{E} - \left(\frac{\partial u}{\partial x}\right)_\mathrm{W} + \left(\frac{\partial u}{\partial y}\right)_\mathrm{N} - \left(\frac{\partial u}{\partial y}\right)_\mathrm{S}\right].$$

Accumulation. Approximate the volume integral by $h^2 u_e$, so its derivative is $h^2\,\mathrm{d}u_e/\mathrm{d}t$. It vanishes for the target steady problem. For backward Euler it becomes $h^2(u_e^{n+1} - u_e^n)/\Delta t$, where n is a physical time index.

3 continued Central interpolation and face gradients

Let $u_{\mathrm{E}}, u_{\mathrm{W}}, u_{\mathrm{N}}, u_{\mathrm{S}}$ be neighboring stored *horizontal* velocities one grid spacing from u_e. They are not the pressure samples p_E, p_W, p_N, p_S. For central interpolation,

$$\widehat{u}_{\mathrm{E}} = (u_e + u_{\mathrm{E}})/2, \qquad \widehat{u}_{\mathrm{W}} = (u_{\mathrm{W}} + u_e)/2,$$
$$\widehat{u}_{\mathrm{N}} = (u_e + u_{\mathrm{N}})/2, \qquad \widehat{u}_{\mathrm{S}} = (u_{\mathrm{S}} + u_e)/2.$$

The advecting velocities are obtained from the staggered grid:

$$c_{\mathrm{E}} = (u_e + u_{\mathrm{E}})/2, \qquad c_{\mathrm{W}} = (u_{\mathrm{W}} + u_e)/2,$$
$$c_{\mathrm{N}} = (v_{n,P} + v_{n,E})/2, \qquad c_{\mathrm{S}} = (v_{s,P} + v_{s,E})/2.$$

For example, $v_{n,P}$ and $v_{n,E}$ are the vertical velocities on the north faces of pressure cells P and E. During a linear momentum solve the c values are held at their current estimates; updating them is a later nonlinear step.

Approximate the gradients at the momentum-volume faces by

$$u_{x,\mathrm{E}} \simeq (u_{\mathrm{E}} - u_e)/h, \qquad u_{x,\mathrm{W}} \simeq (u_e - u_{\mathrm{W}})/h,$$
$$u_{y,\mathrm{N}} \simeq (u_{\mathrm{N}} - u_e)/h, \qquad u_{y,\mathrm{S}} \simeq (u_e - u_{\mathrm{S}})/h.$$

Here $u_{x,\mathrm{E}}$ means $\partial u/\partial x$ evaluated at the east face. Substitution and division by h^2 turn diffusion into

$$\beta(u_{\mathrm{E}} + u_{\mathrm{W}} + u_{\mathrm{N}} + u_{\mathrm{S}} - 4u_e), \qquad \boxed{\beta = \frac{1}{Re\, h^2}.}$$

For instance, the east and west contributions combine as $[(u_{\mathrm{E}} - u_e) - (u_e - u_{\mathrm{W}})]/(Re\, h^2)$. The two occurrences of $-u_e$ explain the central $-2u_e$ in that direction.

The full volume-normalized equation is therefore

$$\begin{aligned}\frac{\mathrm{d}u_e}{\mathrm{d}t} &+ \frac{c_{\mathrm{E}}(u_e + u_{\mathrm{E}}) - c_{\mathrm{W}}(u_{\mathrm{W}} + u_e)}{2h} \\ &+ \frac{c_{\mathrm{N}}(u_e + u_{\mathrm{N}}) - c_{\mathrm{S}}(u_{\mathrm{S}} + u_e)}{2h} \\ &= -\frac{p_E - p_P}{h} + \beta(u_{\mathrm{E}} + u_{\mathrm{W}} + u_{\mathrm{N}} + u_{\mathrm{S}} - 4u_e).\end{aligned}$$

Each term has the form of an acceleration. Dividing face length h by cell area h^2 accounts for the factors of $1/h$ in these terms.

3 continued Momentum coefficients and numerical example

For the steady interior row, move each convective neighbor term to the right and each central diffusive term to the left. The result is

$$\boxed{a_e u_e = a_{\mathrm{E}} u_{\mathrm{E}} + a_{\mathrm{W}} u_{\mathrm{W}} + a_{\mathrm{N}} u_{\mathrm{N}} + a_{\mathrm{S}} u_{\mathrm{S}} + b_e - \frac{p_E - p_P}{h}.}$$

An interior steady row has no time or wall source, so $b_e = 0$ there. The neighbor coefficients are

$$a_{\mathrm{E}} = \beta - c_{\mathrm{E}}/(2h), \qquad a_{\mathrm{W}} = \beta + c_{\mathrm{W}}/(2h),$$
$$a_{\mathrm{N}} = \beta - c_{\mathrm{N}}/(2h), \qquad a_{\mathrm{S}} = \beta + c_{\mathrm{S}}/(2h).$$

For example, the east contribution to convection is $c_{\mathrm{E}} u_e/(2h) + c_{\mathrm{E}} u_{\mathrm{E}}/(2h)$. The east diffusion term on the right is $\beta u_{\mathrm{E}} - \beta u_e$. Moving terms gives $(\beta - c_{\mathrm{E}}/(2h))u_{\mathrm{E}}$ on the right and $(\beta + c_{\mathrm{E}}/(2h))u_e$ on the left. Repeating the operation at all faces gives

$$\begin{aligned} a_e &= 4\beta + \frac{c_{\mathrm{E}} - c_{\mathrm{W}} + c_{\mathrm{N}} - c_{\mathrm{S}}}{2h} \\ &= a_{\mathrm{E}} + a_{\mathrm{W}} + a_{\mathrm{N}} + a_{\mathrm{S}} + \frac{c_{\mathrm{E}} - c_{\mathrm{W}} + c_{\mathrm{N}} - c_{\mathrm{S}}}{h}. \end{aligned}$$

Do not set the last term to zero merely because the target flow is incompressible: the current advecting estimates can still have a discrete mass imbalance. Pressure–velocity coupling removes that imbalance iteratively. Backward Euler adds $1/\Delta t$ to a_e and $u_e^n/\Delta t$ to b_e. Known wall values also contribute to the source and can modify the diagonal.

One frozen-row example. Set $h = 0.25$, $Re = 100$, $c_{\mathrm{E}} = c_{\mathrm{W}} = 0.02$, $c_{\mathrm{N}} = c_{\mathrm{S}} = 0$. Then $\beta = 0.16$ and

$$(a_{\mathrm{E}}, a_{\mathrm{W}}, a_{\mathrm{N}}, a_{\mathrm{S}}) = (0.12, 0.20, 0.16, 0.16), \qquad a_e = 0.64.$$

For given neighbor values $(u_{\mathrm{E}}, u_{\mathrm{W}}, u_{\mathrm{N}}, u_{\mathrm{S}}) = (0.04, 0.02, 0.03, 0.01)$, their weighted sum is

$$0.12(0.04) + 0.20(0.02) + 0.16(0.03) + 0.16(0.01) = 0.0152.$$

With $p_P = 0.010$, $p_E = 0.008$ and $b_e = 0$, the pressure contribution is $-(0.008 - 0.010)/0.25 = +0.008$. Thus

$$\boxed{u_e = (0.0152 + 0.008)/0.64 = 0.03625.}$$

The lower east pressure drives a positive horizontal contribution. This satisfies one frozen momentum row; it does not by itself enforce the neighboring pressure cells' continuity equations. That is the purpose of SIMPLE. For vertical momentum, apply the same construction around v_n; its pressure term is $-(p_N - p_P)/h$. The next section assembles these scalar rows into a matrix.

4 Momentum coefficients and matrix notation

For a velocity location labeled f, let $\mathrm{NB}(f)$ be the neighboring velocity locations in its momentum row. The index g labels one such location; a_{fg} is the coefficient multiplying its velocity q_g. These are velocity-control-volume neighbors, rather than the pressure neighbors labeled E, W, N, S. Collect the source and neighbor contributions into

$$H_f = b_f + \sum_{g \in \mathrm{NB}(f)} a_{fg} q_g, \qquad d_f = \frac{1}{a_f}.$$

The symbol q_g means a velocity degree of freedom; it is not volume flux. The scalar momentum row becomes

$$q_f = d_f \left[H_f - (Gp)_f \right], \qquad (Gp)_e = \frac{p_E - p_P}{h}.$$

The vector q collects all unknown interior horizontal and vertical face velocities. The scalar rows form two component linear systems, written together as

$$Aq = b - Gp.$$

The product Dq collects the cell residuals: $(Dq)_P = r_{c,P}$. The product Gp collects face pressure gradients. Multiplication by d scales each face entry by its own $1/a_f$.

For N cells per side, p has N^2 entries and q has $2N(N-1)$ unknown interior face velocities. Thus A is $2N(N-1)$ square, D has N^2 rows and $2N(N-1)$ columns, G has the reverse shape, and $L = -DdG$ is N^2 square. The operator $\mathrm{diag}(a_f)$ places the coefficients on the diagonal and zeros elsewhere.

The diagonal of A contains the center coefficients a_f; its off-diagonal entries are $-a_{fg}$. With $R = A - \mathrm{diag}(a_f)$ and $d = \mathrm{diag}(1/a_f)$,

$$H(q) = b - Rq, \qquad q = d\left[H(q) - Gp\right].$$

Thus H is computed from current neighbor velocities; it is not another fluid property. Here d is an inverse diagonal matrix, while h is mesh spacing. In operator formulas L denotes the pressure matrix; the L used earlier in dimensional scaling denotes cavity length.

Symbol	Meaning	Location or role
p	Pressure relative to a reference	Cell centers
q	Collected velocity components	Interior faces
A, b	Momentum matrix and source	Two component systems
d	Inverse center coefficients	Face-wise diagonal
R	Off-diagonal part of A	Neighbor coupling
$H = b - Rq$	Source plus neighbor terms	Recomputed from q
D	Discrete divergence	Faces to cell balance
G	Discrete pressure gradient	Centers to faces
$L = -DdG$	Pressure operator	Cell-center system

The notation abbreviates the scalar equations; it does not introduce additional physics. Each occurrence of A, b, d refers to the current assembly, including any wall, transient or relaxation modifications.

5 Pressure correction from the momentum equations

A star denotes a tentative velocity. The superscript k labels the current steady coupling iteration, not a power or a physical time level. Given the current pressure estimate p^k, solve momentum for q^* while holding its coefficients fixed:

$$Aq^* = b - Gp^k.$$

Pressure correction and pressure differences. Let p_{corr} be the pressure targeted by the current correction solve. Define

$$p' = p_{\text{corr}} - p^k, \qquad q' = q_{\text{corr}} - q^*.$$

A prime is an additive correction, not a derivative. The unrelaxed pressure change at that cell is

$$\boxed{\Delta p = p_{\text{corr}} - p^k = p'.}$$

It is a change at a fixed cell during coupling. By contrast, $p_E - p_P$ is a spatial pressure difference between neighboring cells at the same iteration. Neither is the continuity residual $r_{c,P}$.

Write $q_{\text{corr}} = q^* + q'$ and $p_{\text{corr}} = p^k + p'$. Substituting these into the same momentum system and subtracting the predictor gives

$$Aq' = -Gp'.$$

At one horizontal face this is

$$a_e u'_e = \sum_{g \in \text{NB}(e)} a_{eg} q'_g - \frac{p'_E - p'_P}{h}.$$

The exact correction couples neighboring velocity corrections. SIMPLE obtains a local correction by neglecting that neighbor-correction sum during this correction solve:

$$\boxed{u'_e = -d_e \frac{p'_E - p'_P}{h}}, \qquad d_e = 1/a_e.$$

Likewise,

$$u'_w = -d_w \frac{p'_P - p'_W}{h},$$
$$v'_n = -d_n \frac{p'_N - p'_P}{h}, \qquad v'_s = -d_s \frac{p'_P - p'_S}{h}.$$

The corrected velocity must satisfy cell continuity. Substituting the four face corrections gives

$$0 = r^*_{c,P} + \frac{u'_e - u'_w + v'_n - v'_s}{h}, \qquad r^*_{c,P} = \frac{u^*_e - u^*_w + v^*_n - v^*_s}{h}.$$

This equation determines pressure corrections from the predictor's cell imbalance. Neglecting neighbor corrections is an approximation to momentum coupling, not permission to violate continuity. Repeated coupling iterations are required to recover the target steady momentum solution.

6 The cell pressure equation and its boundary data

At each face, the corrected velocity is tentative velocity plus its correction:

$$u_e = u_e^* - \frac{d_e}{h}(p'_E - p'_P), \qquad u_w = u_w^* - \frac{d_w}{h}(p'_P - p'_W),$$
$$v_n = v_n^* - \frac{d_n}{h}(p'_N - p'_P), \qquad v_s = v_s^* - \frac{d_s}{h}(p'_P - p'_S).$$

Insert these into $(u_e - u_w + v_n - v_s)/h = 0$. The west and south terms change sign because their velocities enter cell outflow with minus signs:

$$\begin{aligned} 0 = r^*_{c,P} &- \frac{d_e}{h^2}(p'_E - p'_P) + \frac{d_w}{h^2}(p'_P - p'_W) \\ &- \frac{d_n}{h^2}(p'_N - p'_P) + \frac{d_s}{h^2}(p'_P - p'_S). \end{aligned}$$

Collect the coefficients of each pressure correction and move the tentative residual to the right:

$$\begin{aligned} &\frac{d_e + d_w + d_n + d_s}{h^2} p'_P \\ &- \frac{d_e p'_E + d_w p'_W + d_n p'_N + d_s p'_S}{h^2} = -r^*_{c,P}. \end{aligned}$$

The center coefficient is the sum of four face weights; each neighbor has a negative coefficient. The right side is the negative cell divergence of the tentative velocity. The same face weight must be shared by its two adjacent pressure cells.

Matrix form. Applying D to $q = q^* - dGp'$ gives

$$\begin{aligned} 0 &= Dq^* - DdGp', \\ L &= -DdG, \\ Lp' &= -Dq^*. \end{aligned}$$

The positive-diagonal convention for L explains the minus sign on the right. Once pressure is solved, the face correction $-dGp'$ cancels the computed cell imbalance, up to linear-solve error.

6 continued Boundary cells and pressure reference

Impermeable walls. The normal wall velocity is prescribed and is not an unknown interior face velocity. Its normal correction is zero. A prescribed wall-normal face contributes zero correction weight, so that face coefficient is omitted from the boundary-cell pressure row. Pressure assembly therefore has no correction flux across the external wall. For a cell adjacent to a stationary west wall, $u^*_w = u'_w = 0$, so its row is

$$\frac{d_e + d_n + d_s}{h^2} p'_P - \frac{d_e p'_E + d_n p'_N + d_s p'_S}{h^2} = -r^*_{c,P}.$$

There is no west pressure-neighbor term. At a corner, omit both prescribed wall-normal face weights. Zero normal correction does not mean zero pressure at every wall.

Pressure reference. The vector $\mathbf{1}$ contains one in every pressure entry. Adding a constant to every cell pressure changes no pressure gradient: $G\mathbf{1} = 0$. Consequently $L\mathbf{1} = 0$ and the closed-cavity pressure system has one arbitrary constant. Pinning means temporarily prescribing one pressure unknown, for example $p_{\text{ref cell}} = 0$, to make the linear solve unique. The implementation then removes the arithmetic cell mean:

$$\bar{p} = \frac{1}{N^2} \sum_P p_P, \qquad \widetilde{p}_P = p_P - \bar{p}.$$

The same constant is subtracted from every cell, so

$$\widetilde{p}_E - \widetilde{p}_P = (p_E - \bar{p}) - (p_P - \bar{p}) = p_E - p_P.$$

The gradient and corrected velocity are unchanged. The reported zero-mean pressure is a gauge choice, not an extra boundary force.

Tangential no-slip is enforced in momentum through wall diffusion. On the staggered grid it is distinct from the zero normal correction used in the cell mass balance.

7 Two-cell pressure-correction example

Consider two closed square cells with one shared vertical face. This is an algebraic conservation example, not a resolved cavity. The pressure guess is initially zero in both cells. Let $h = 0.5$, tentative shared velocity $u^* = 0.2$, and inverse momentum diagonal $d_e = 0.1$. Positive shared velocity leaves cell 1 and enters cell 2. All external normal velocities are zero.

$$r^*_{c,1} = \frac{0.2}{0.5} = 0.4, \qquad r^*_{c,2} = \frac{-0.2}{0.5} = -0.4.$$

The shared pressure weight is $d_e/h^2 = 0.1/0.25 = 0.4$. Therefore

$$0.4 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} p'_1 \\ p'_2 \end{bmatrix} = \begin{bmatrix} -0.4 \\ 0.4 \end{bmatrix}.$$

The first row gives $p'_1 - p'_2 = -1$. With zero mean, $p'_1 + p'_2 = 0$, so $p'_1 = -0.5$ and $p'_2 = 0.5$. The corrected face velocity is

$$u = u^* - d_e \frac{p'_2 - p'_1}{h} = 0.2 - 0.1 \frac{0.5 - (-0.5)}{0.5} = 0.$$

Thus the unrelaxed changes are $\Delta p_1 = p'_1 = -0.5$ and $\Delta p_2 = p'_2 = 0.5$. Both corrected divergences are zero. A common constant added to p'_1 and p'_2 leaves this calculation unchanged.

Pressure relaxation and velocity correction. The pressure relaxation factor α_p is the fraction of p' added to the pressure stored for the next momentum predictor: $p^{k+1} = p^k + \alpha_p p'$. The stored pressure change is therefore $\alpha_p p'$, rather than the full unrelaxed $\Delta p = p'$. If only $\alpha_p = 0.3$ of the velocity correction were applied, the shared velocity would remain $0.2 - 0.3(0.2) = 0.14$. The two mass defects would then be $+0.28$ and -0.28. With $\alpha_p = 0.3$, the stored pressures become -0.15 and 0.15 from the zero initial guess, while the fully corrected shared velocity remains zero. This velocity need not yet satisfy momentum with the relaxed stored pressure; the next predictor and correction continue the coupling iteration. The conservative velocity correction is applied fully.

Local and global conservation. The initial global sum was already $0.4 - 0.4 = 0$, but each cell was imbalanced. A zero total boundary flux alone cannot certify local continuity.

8 SIMPLE: steady pressure–velocity coupling

SIMPLE means Semi-Implicit Method for Pressure-Linked Equations. Its index k counts steady coupling iterations, not physical time. Each iteration assembles convection from the latest velocity, solves a momentum predictor, then solves the pressure-correction equation.

Momentum under-relaxation. Let a_f^0 be the original center coefficient of the current unrelaxed assembly. The superscript 0 here identifies the coefficient before relaxation, not the initial physical time. The label rel means relaxed. The factor α_u controls momentum under-relaxation; for $0 < \alpha_u \leq 1$, the relaxed row uses

$$a_f^{\text{rel}} = \frac{a_f^0}{\alpha_u}, \qquad b_f^{\text{rel}} = b_f + \left(\frac{1}{\alpha_u} - 1\right) a_f^0 q_f^k.$$

The added source uses the original diagonal a_f^0. At a converged fixed point, $q_f^{k+1} = q_f^k$, the added diagonal and source contributions cancel. The target steady equation is unchanged. The pressure correction uses $d_f = 1/a_f^{\text{rel}}$.

One steady iteration.

$$\begin{aligned} A^{\text{rel}} q^* &= b^{\text{rel}} - G p^k, \\ L p' &= -D q^*, \\ q^{k+1} &= q^* - d G p', \\ p^{k+1} &= p^k + \alpha_p p', \qquad 0 < \alpha_p \leq 1. \end{aligned}$$

Velocity is corrected fully; only the stored pressure update is damped by α_p. After boundary conditions and the pressure gauge are applied, the next iteration rebuilds the momentum coefficients from the corrected velocity.

Stopping criteria. Momentum relaxation can make updates small before the original equations are satisfied. The implementation checks the unrelaxed nonlinear steady momentum defect and the cell divergence. Both must meet tolerance. In the retained baseline, $\alpha_u = 0.7$ and $\alpha_p = 0.3$.

Role. SIMPLE is a natural steady solver. It can also be used inside a transient nonlinear solve, but the current implementation's SIMPLE branch has no physical time step. Its iteration count cannot be compared directly with the physical-step counts of PISO or PIMPLE.

9 Physical time and backward-Euler momentum

PISO and PIMPLE solve a transient momentum equation. Superscripts n and $n+1$ denote physical time levels, separated by Δt. For one face velocity,

$$\left.\frac{\partial q_f}{\partial t}\right|_{n+1} \approx \frac{q_f^{n+1} - q_f^n}{\Delta t}.$$

If the spatial momentum row has center coefficient a_f^{space} and source b_f^{space}, backward Euler produces

$$\left(a_f^{\text{space}} + \frac{1}{\Delta t}\right) q_f^{n+1} = \sum_g a_{fg} q_g^{n+1} + b_f^{\text{space}} + \frac{q_f^n}{\Delta t} - (Gp^{n+1})_f.$$

Thus the transient assembly adds $1/\Delta t$ to the diagonal and the known old velocity $q_f^n/\Delta t$ to the source.

Old-time velocity is fixed. At the beginning of the physical step, store q^n. Every momentum assembly and every correction within that step uses the same old-time source. Replacing q^n with an intermediate corrected velocity would change the discrete physical problem.

Nonlinear coefficients. Convective coefficients depend on advecting velocity. They must be evaluated from a known estimate to form a linear predictor system. Such coefficients are called frozen or lagged during that solve. A perfectly solved linear predictor can still differ from the fully implicit nonlinear step if its coefficients were evaluated at an outdated velocity.

Three iteration levels. A linear solver solves one matrix equation. A pressure corrector couples pressure and velocity for a frozen momentum matrix. An outer corrector rebuilds that matrix using the latest velocity. None of these is an additional physical time advance.

Backward Euler is first-order accurate in time. Extra correctors can reduce algebraic coupling error, but do not change its time-discretization order. A larger stable Δt still requires a temporal-accuracy assessment.

10 PISO: pressure corrections for a frozen momentum matrix

PISO means Pressure-Implicit with Splitting of Operators. At one physical step, assemble transient momentum once and solve

$$Aq^{*} = b - Gp^{\text{guess}}.$$

This predictor uses the pressure available from the preceding step or initial state. Its momentum residual for that pressure can be small, while its cell divergence is nonzero.

Set $q^{(0)} = q^{*}$. The index m counts inner corrections within this one physical step. The component matrices A, their source b, inverse diagonal d, off-diagonal part R, and pressure operator $L = -DdG$ remain fixed throughout the inner loop. Neighbor velocities do change, so each correction computes

$$\begin{aligned} H^{(m)} &= b - Rq^{(m)}, \\ Lp^{(m+1)} &= -D\left(dH^{(m)}\right), \\ q^{(m+1)} &= d\left[H^{(m)} - Gp^{(m+1)}\right]. \end{aligned}$$

The pressure equation follows by applying D to the last line and imposing $Dq^{(m+1)} = 0$. Unlike SIMPLE's p', this implementation's $p^{(m+1)}$ is the full pressure field in the chosen gauge, rather than a pressure increment. It replaces the previous pressure; it is not added to it.

Purpose of the next correction. Reconstructed velocity changes the neighbor terms in H. Recomputing H and solving pressure again improves the consistency of reconstructed velocity with the same frozen momentum equation. Repeating a linear pressure solve with an unchanged right-hand side is not an additional PISO correction.

Let N_i be the specified number of inner corrections. After N_i corrections, accept $q^{n+1} = q^{(N_i)}$ and $p^{n+1} = p^{(N_i)}$, then advance time once. The retained baseline has $N_i = 2$. No momentum or pressure relaxation is applied by this transient branch.

The nonlinear convection coefficients remain those used for the initial predictor. That limitation motivates an outer coefficient-update loop.

11 PISO: a frozen-matrix numerical example

This reduced two-velocity system illustrates coupling algebra; it is not a cavity grid. Let

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & -1 \end{bmatrix}, \quad G = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Here p is one independent scalar pressure variable after removing an arbitrary common pressure offset. It is not a separate pressure value in each physical cell. The matrices are prescribed to illustrate coupling algebra; they are not derived from the earlier two-cell geometry. Continuity is $Dq = q_1 - q_2 = 0$. From A,

$$d = \begin{bmatrix} 1/4 & 0 \\ 0 & 1/3 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad L = -DdG = 7/12.$$

Starting from $p^{\text{guess}} = 0$, the momentum predictor solves $Aq^* = b$:

$$q^* = \begin{bmatrix} 6/11 \\ -2/11 \end{bmatrix}, \qquad Dq^* = 8/11.$$

It satisfies momentum exactly but violates continuity. The first inner correction uses

$$H^{(0)} = b - Rq^* = \begin{bmatrix} 24/11 \\ -6/11 \end{bmatrix}, \qquad dH^{(0)} = \begin{bmatrix} 6/11 \\ -2/11 \end{bmatrix}.$$

Solve pressure and reconstruct velocity:

$$\frac{7}{12}p^{(1)} = -D(dH^{(0)}) = -\frac{8}{11}, \qquad p^{(1)} = -\frac{96}{77},$$
$$q_1^{(1)} = \frac{1}{4}\left(\frac{24}{11} + p^{(1)}\right) = \frac{18}{77}, \qquad q_2^{(1)} = \frac{1}{3}\left(-\frac{6}{11} - p^{(1)}\right) = \frac{18}{77}.$$

The signs in these two velocity formulas follow from $G = (-1, 1)^T$. Now $Dq^{(1)} = 0$. Nevertheless, substituting these fields into the original frozen momentum equation gives

$$r^{(1)} = Aq^{(1)} - b + Gp^{(1)} = \begin{bmatrix} 32/77 \\ -24/77 \end{bmatrix}.$$

The vector $r = Aq - b + Gp$ is the error in satisfying the momentum rows; exact satisfaction gives $r = 0$. Its infinity norm is the largest absolute entry, $\|r\|_\infty = \max_i |r_i|$, where i labels a momentum row. Mass balance is exact here, but momentum consistency still needs improvement.

11 continued The second pressure correction

The corrected neighbor velocities change H, even though A and L are unchanged:

$$H^{(1)} = b - Rq^{(1)} = \begin{bmatrix} 136/77 \\ -18/77 \end{bmatrix}, \qquad dH^{(1)} = \begin{bmatrix} 34/77 \\ -6/77 \end{bmatrix}.$$

The next pressure and velocity are

$$\frac{7}{12}p^{(2)} = -\frac{40}{77}, \qquad p^{(2)} = -\frac{480}{539},$$
$$q^{(2)} = d\left[H^{(1)} - Gp^{(2)}\right] = \begin{bmatrix} 118/539 \\ 118/539 \end{bmatrix}.$$

The pressure is replaced by $p^{(2)}$; adding $p^{(2)}$ to $p^{(1)}$ would be incorrect. Again continuity is exact. The remaining frozen-momentum defect is

$$r^{(2)} = \begin{bmatrix} -8/539 \\ -8/539 \end{bmatrix}.$$

Stage	$\lvert Dq\rvert$	$\lVert Aq - b + Gp\rVert_\infty$
Momentum predictor	8/11	0
Inner correction 1	0	$32/77 \approx 0.41558$
Inner correction 2	0	$8/539 \approx 0.01484$

The exact constrained solution is $q_1 = q_2 = 2/9$ and $p = -8/9$. It follows by imposing $q_1 = q_2 = s$ in both original momentum rows: $5s - p = 2$ and $4s + p = 0$, hence $9s = 2$.

The second correction reduces momentum inconsistency in this example while preserving mass balance. This is a demonstrated property of these specified matrices, not a guarantee of monotone convergence for every CFD system.

The right-hand side of the pressure equation changes from –8/11 to –40/77. The example therefore distinguishes another PISO correction from another iteration of an unchanged linear pressure solve.

12 PIMPLE: outer coefficient updates and inner corrections

PIMPLE combines an outer momentum-update loop with PISO inner corrections. Physical time levels are $n, n+1$; ℓ labels outer updates and m labels inner pressure corrections. The previous physical velocity q^n is stored once.

At the start of outer pass ℓ, assemble convection using the latest available corrected velocity $\widehat{q}^{(\ell)}$. With I denoting the identity matrix, this gives new matrices and sources

$$A^{(\ell)} = A_{\text{space}}(\widehat{q}^{(\ell)}) + I/\Delta t, \qquad b^{(\ell)} = b^{(\ell)}_{\text{wall}} + q^n/\Delta t.$$

The same q^n appears in every outer pass. Wall contributions may depend on the current assembly, but the prescribed wall motion and old-time source belong to the same physical step.

The first outer pass starts from q^n, p^n. The symbol $\widehat{p}^{(\ell)}$ denotes the latest available pressure at the start of outer pass ℓ. A new momentum predictor is solved using that pressure. Then inner corrections hold the current outer assembly fixed:

$$\begin{aligned} A^{(\ell)} q^{(\ell,0)} &= b^{(\ell)} - G\widehat{p}^{(\ell)}, \\ H^{(\ell,m)} &= b^{(\ell)} - R^{(\ell)} q^{(\ell,m)}, \\ L^{(\ell)} p^{(\ell,m+1)} &= -D\left(d^{(\ell)} H^{(\ell,m)}\right), \\ q^{(\ell,m+1)} &= d^{(\ell)}\left[H^{(\ell,m)} - G p^{(\ell,m+1)}\right]. \end{aligned}$$

With N_i inner corrections per outer pass, the initial and transfer rules are

$$\widehat{q}^{(1)} = q^n, \qquad \widehat{p}^{(1)} = p^n,$$
$$\widehat{q}^{(\ell+1)} = q^{(\ell,N_i)}, \qquad \widehat{p}^{(\ell+1)} = p^{(\ell,N_i)}.$$

Thus the final corrected fields of one outer pass initialize the next. Rebuilding A also rebuilds d, R and L. Each inner solve computes the full pressure field in the chosen gauge, as in this PISO implementation.

Two different purposes. Inner corrections improve pressure–velocity consistency for the current matrix. Outer passes update the nonlinear coefficients that matrix represents. Neither loop advances physical time. Only after the final outer pass are the accepted fields assigned to time $n+1$.

With one outer pass, this code reduces to its PISO algorithm. Its PIMPLE branch does not apply SIMPLE's under-relaxation. Additional loops reduce certain algebraic errors at extra cost; they do not increase backward-Euler's formal time order.

13 PIMPLE: a nonlinear coefficient example

Extend the preceding algebraic example by making its momentum coefficients depend on velocity:

$$A(s) = \begin{bmatrix} 4+s & 1 \\ 1 & 3+s \end{bmatrix}, \qquad s = (q_1 + q_2)/2, \qquad b = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

Keep the same D, G and initial $q = 0, p = 0$. At each outer pass, evaluate s from its starting velocity, freeze $A(s)$, solve a new predictor, and perform two inner corrections. The source b remains fixed. This toy coefficient law illustrates lagging; it is not a Navier–Stokes convection discretization.

At outer pass 1, $s = 0$, so the two inner corrections are exactly the previous PISO example. They end at $q_1 = q_2 = 0.2189239332$. The next outer pass uses

$$A^{(2)} = \begin{bmatrix} 4.2189239332 & 1 \\ 1 & 3.2189239332 \end{bmatrix},$$

and recomputes d, R, L before the new predictor and inner corrections.

Outer pass	Frozen s	Final $q_1 = q_2$	Final p	Nonlinear defect
1	0	0.2189239332	–0.8905380334	3.30854×10^{-2}
2	0.2189239332	0.2119026446	–0.8940486777	1.53537×10^{-3}
3	0.2119026446	0.2122289655	–0.8938855173	7.14784×10^{-5}

The reported defect is the largest absolute component of the nonlinear momentum residual, evaluated with coefficients rebuilt from the final velocity:

$$\| A\left((q_1 + q_2)/2\right) q - b + Gp \|_\infty .$$

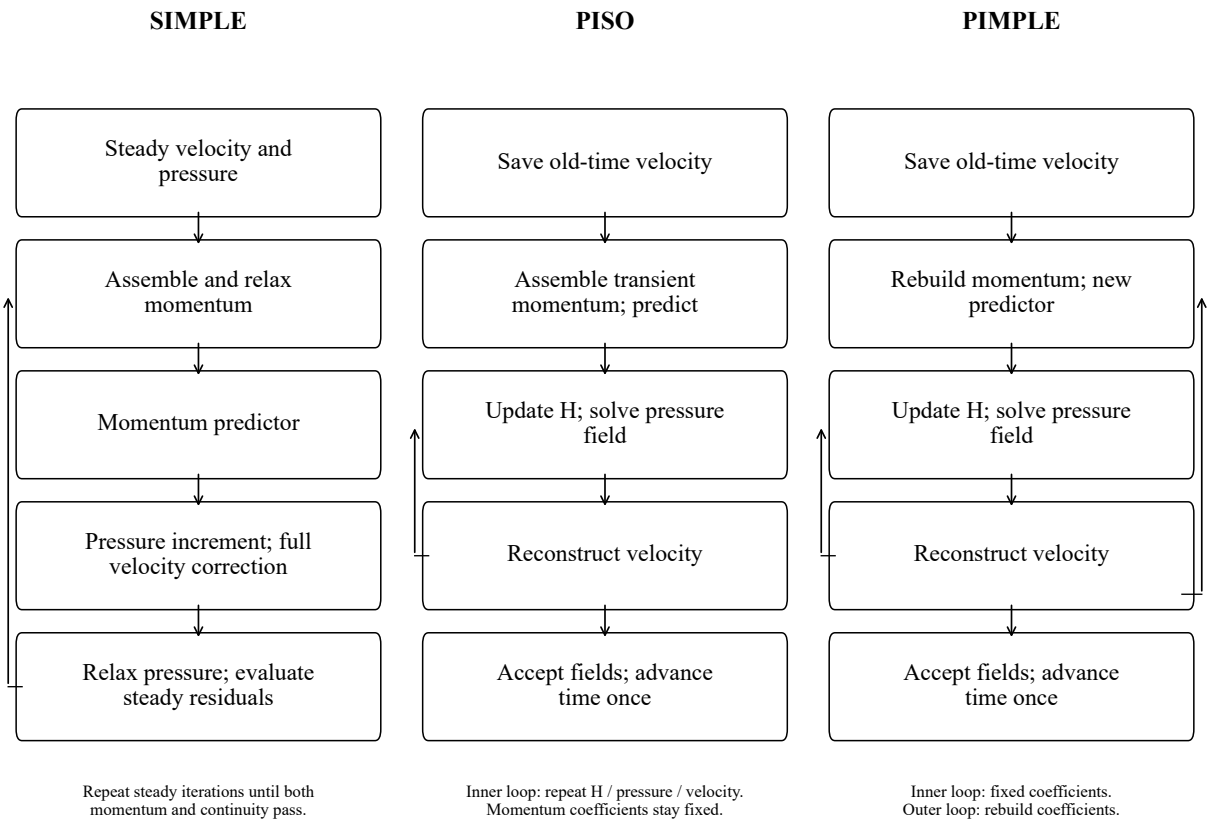
It is different from the residual of the frozen predictor matrix. All final mass defects are below 10^{-15}.

For the exact constrained solution set $q_1 = q_2 = s$ and add the two momentum rows:

$$(9 + 2s)s = 2, \qquad s = \frac{\sqrt{97} - 9}{4} = 0.2122144504, \qquad p = s/2 - 1.$$

The table shows the benefit of coefficient updates for this example. Actual cavity efficiency must be judged using its own measured equation defects, accuracy and solve costs.

14 Algorithm loops and computational work



The nested loops act on one target problem at the current physical step. The diagram distinguishes steady SIMPLE iterations from PISO and PIMPLE time integration.

Let N_i be the number of pressure correctors and N_o the number of outer correctors. Each predictor solves horizontal and vertical momentum separately.

Work unit	SIMPLE iteration	PISO step	PIMPLE step
Momentum assemblies	1	1	N_o
Component momentum solves	2	2	$2N_o$
Pressure solves	1	N_i	N_oN_i
Physical time advances	0	1	1

For three outer and two inner correctors, PIMPLE uses three assemblies, six component momentum solves and six pressure solves per physical step. A direct linear solver may factor a pressure matrix once and reuse it within an inner loop, while solving different right-hand sides.

An iteration count alone cannot compare these costs. The retained runs use fixed corrector counts, rather than an adaptive inner tolerance. A steady stopping test is applied outside these operations.

15 Convergence, accuracy and the meaning of a residual

Cell continuity. The mass-balance diagnostic is $\max_P |r_{c,P}|$: the largest absolute continuity residual among all pressure cells. It is evaluated from face velocities, not from an interpolated picture of cell-center velocity.

Momentum defect. For a steady target, the implementation rebuilds the unrelaxed spatial coefficients using final velocity and evaluates the original momentum equations. For a transient step, a nonlinear step defect includes the same fixed old-time source:

$$r_{\text{step}} = A_{\text{space}}(q^{n+1})q^{n+1} - b_{\text{wall}} + \frac{q^{n+1} - q^n}{\Delta t} + Gp^{n+1}.$$

Small mass defect alone does not make this vector small, as the PISO numerical example demonstrated.

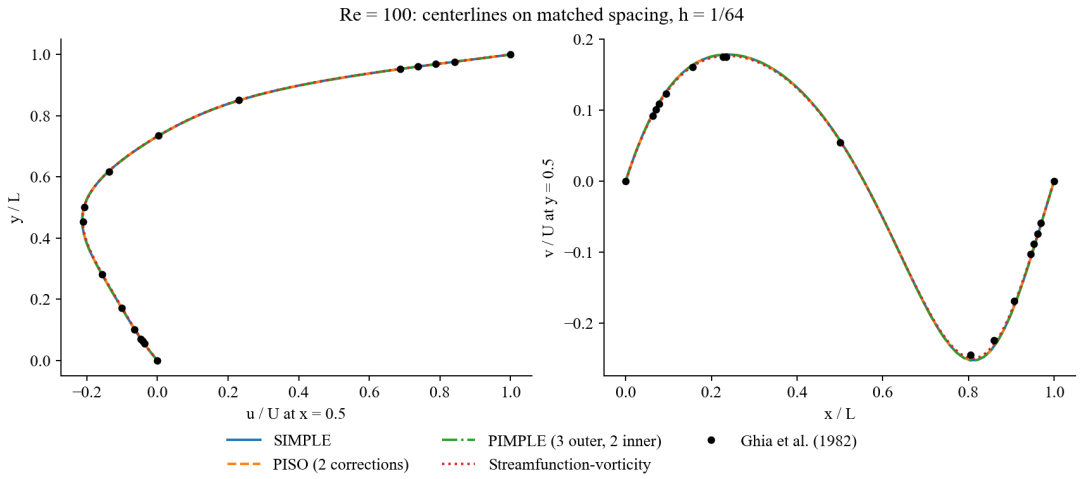
Accuracy. Agreement with independent reference velocities and stability under spatial refinement address solution accuracy. Ghia centerline tables validate selected velocity profiles; they do not validate pressure. A converged coarse-grid calculation can still be inaccurate.

Transient accuracy. Corrector loops reduce algebraic errors inside a step. Temporal error is assessed separately by reducing Δt and comparing at matched physical times. Steady cavity agreement alone does not validate startup or oscillatory flow.

Streamfunction–vorticity comparison. Week 1 advances vorticity and solves streamfunction. Its native update norm is different from the finite-volume momentum defect. Their convergence histories use separate axes; common velocity errors and wall-clock times provide cross-formulation comparisons. N finite-volume cells correspond to $N+1$ finite-difference nodes at matching spacing $h = 1/N$.

Higher Reynolds numbers. Stronger convection and thinner layers can require more cells and more work. More pressure corrections cannot recover spatial structures missing from the mesh. Refinement and higher-*Re* runs can take substantially longer and need more memory; the supplied coarse cases are the starting point for laptop or Colab use.

16 Recorded cavity accuracy and runtime at $Re = 100$



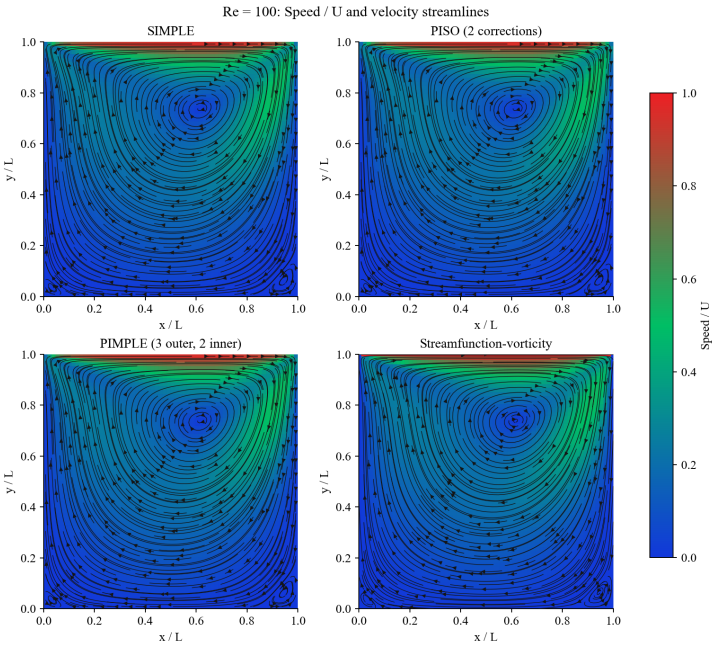
The Ghia velocity markers are independent reference samples. The three finite-volume coupling methods use the same spatial equations. Their converged fields agree closely; differences from the finite-difference formulation also include spatial-discretization differences. The displayed 32/64 finite-volume cell counts correspond to 33/65 finite-difference nodes.

Formulation	Cells per side	Iterations / steps	Recorded seconds
SIMPLE	32	480	5.074
PISO	32	540	6.994
PIMPLE	32	540	19.604
Streamfunction–vorticity	32	19000	8.106
SIMPLE	64	1870	137.987
PISO	64	540	42.756
PIMPLE	64	540	146.522
Streamfunction–vorticity	64	19000	17.284

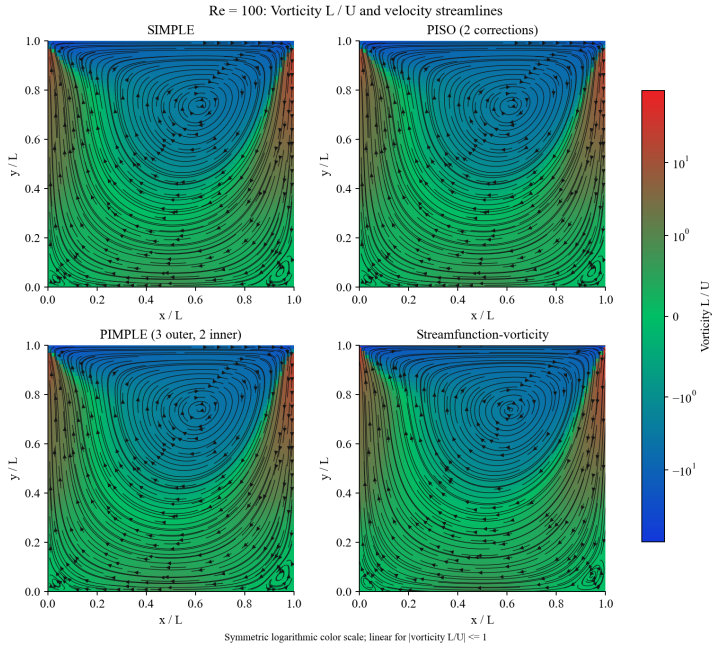
The table includes all four formulations and reports recorded solver-call seconds. The finite-difference reference has its own stopping norm and recovers pressure afterward. These measurements depend on the host, implementation, mesh, relaxation and stopping targets; they do not establish a universal fastest method.

The finite-difference centerline errors decrease on refinement in this retained comparison. The finite-volume Ghia errors are not monotone between the two supplied grids. That behavior is retained rather than interpreted as proof of grid independence. More resolution and reference evidence are needed for an asymptotic accuracy claim.

17 Velocity structure and vorticity

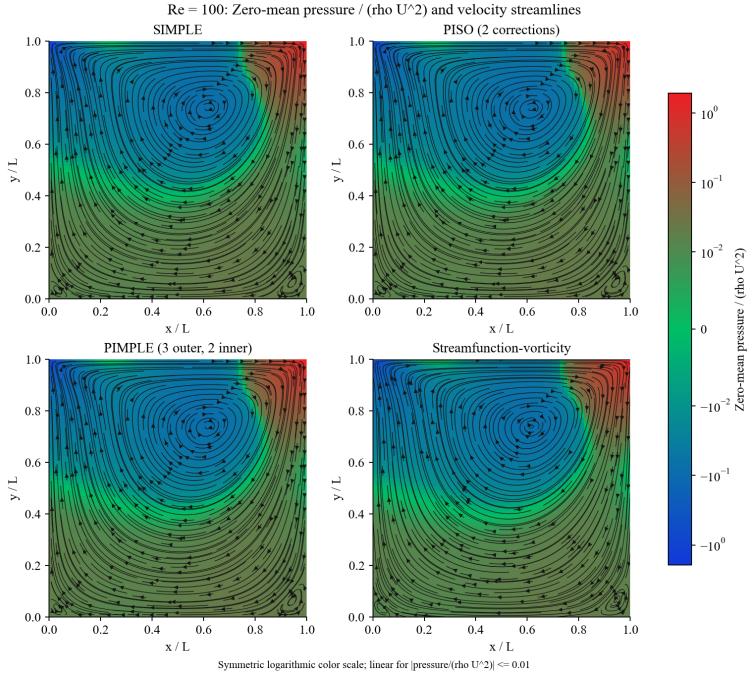


Speed is $|\boldsymbol{u}| = \sqrt{u^2 + v^2}$ and signed vorticity is $\omega = \partial v/\partial x - \partial u/\partial y$. The moving lid drives a clockwise primary circulation and a lower-cavity return flow. Black streamlines follow computed velocity; they are not vorticity contours. Small corner eddies depend on resolution and convergence as well as display density.

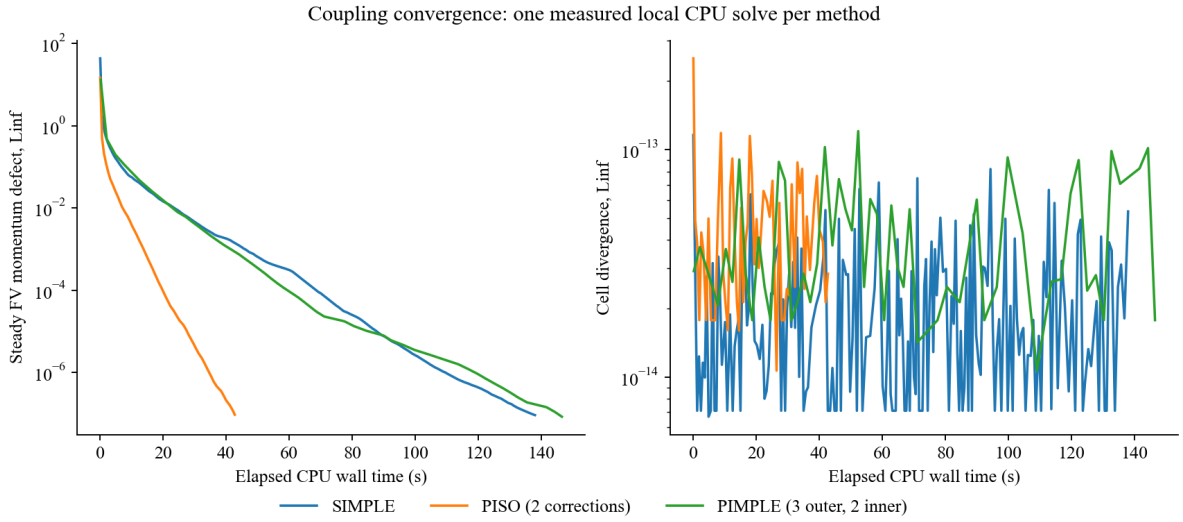


Signed fields use a common symmetric-logarithmic color scale across methods, resolving interior values while retaining strong wall gradients. Colorbar values remain physical nondimensional field values. Similar converged fields are expected when the same discrete steady equations are solved; the comparison is useful for checking consistency.

18 Pressure and native convergence histories

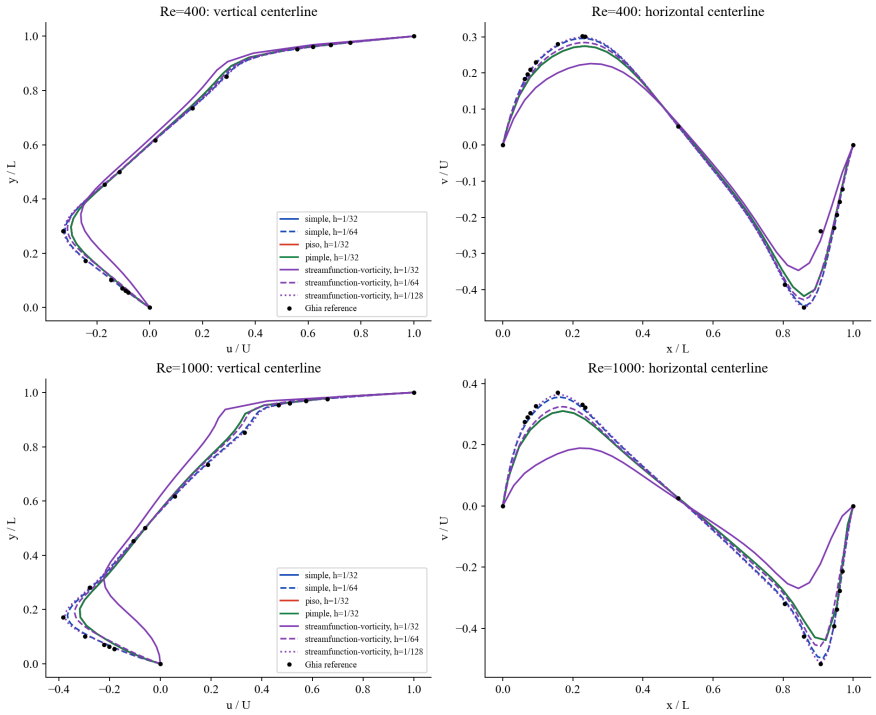


All pressures are reported with zero mean. Pressure gradients, rather than the arbitrary mean, affect velocity. Differences near the discontinuous lid corners should not be interpreted as reference-pressure validation. Ghia supplies velocity data, not pressure data.



The finite-volume histories evaluate momentum and mass balance. The streamfunction solver's native vorticity-update norm is kept separate in the notebook because it is a different quantity. Timing comparisons still include that formulation. A small relaxed update alone is not a momentum-equation convergence test.

19 Accuracy and recorded cost at $Re = 400$ and 1000

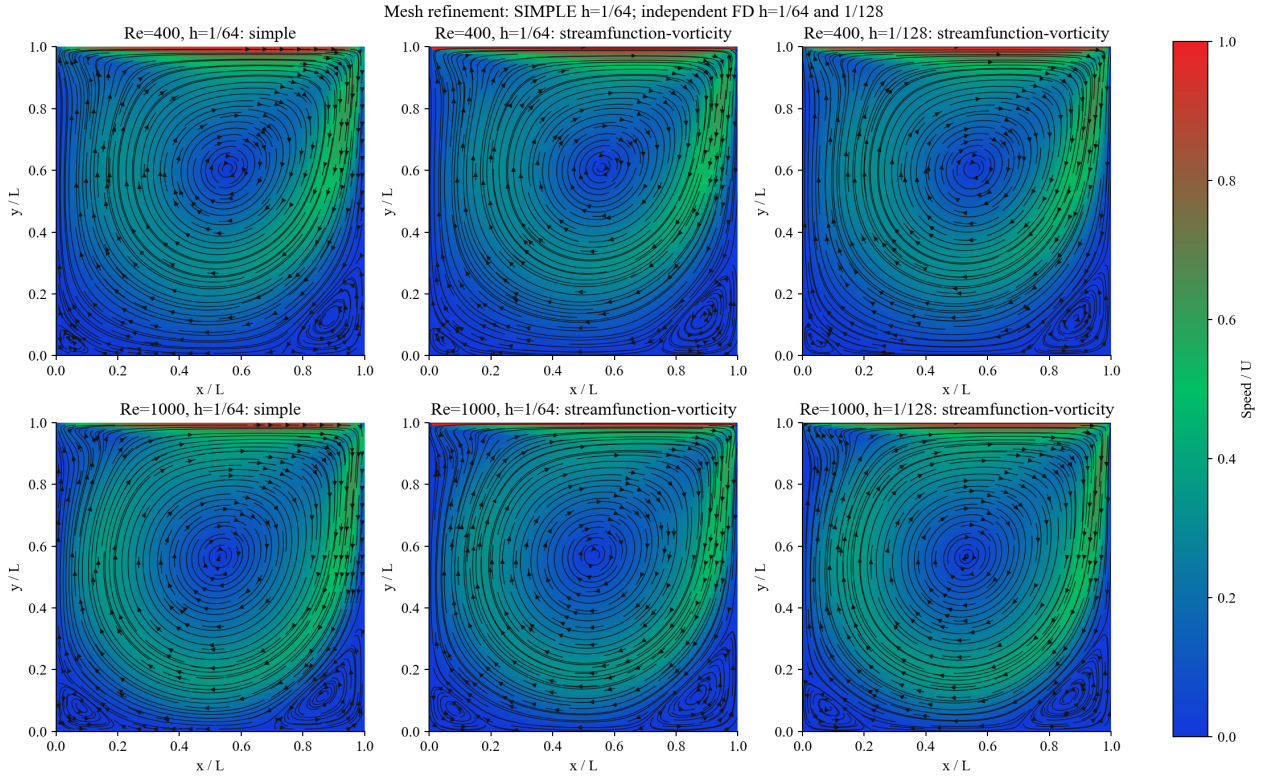


The retained study includes all four methods at $h = 1/32$, plus selected finer meshes. A converged coarse-grid streamfunction result can have large reference error. At $Re = 1000$, its 128-cell-spacing relative profile errors are approximately 2.0% for u and 1.7% for v, using $\|f_{\text{computed}} - f_{\text{Ghia}}\|_2 / \|f_{\text{Ghia}}\|_2$ at the tabulated samples.

Formulation	Re	Cells per side	Recorded seconds
PIMPLE	400	32	96.949
PISO	400	32	22.278
SIMPLE	400	32	11.793
Streamfunction–vorticity	400	32	78.475
SIMPLE	400	64	71.061
Streamfunction–vorticity	400	64	28.382
Streamfunction–vorticity	400	128	70.928
PIMPLE	1000	32	175.140
PISO	1000	32	55.302
SIMPLE	1000	32	12.459
Streamfunction–vorticity	1000	32	175.065
SIMPLE	1000	64	62.141
Streamfunction–vorticity	1000	64	69.579
Streamfunction–vorticity	1000	128	171.387

For an error vector z over the reference samples, $\|z\|_2 = \sqrt{\sum_j z_j^2}$; j labels a sampled location. These are recorded sequential local solver-call timings. Reference errors, mesh spacing and native stopping definitions accompany them. The $Re = 400$ all-point v error retains the isolated published entry; a separately labeled sensitivity metric omits that sample without replacing its value.

20 Mesh refinement and recirculation



Each panel states Reynolds number, formulation and spacing. The sampled primary-vortex location and resolved corner circulation change with refinement. More inner pressure corrections cannot reproduce an unresolved boundary layer or secondary vortex.

The fine-grid study is selected rather than a complete four-method refinement matrix. A result at one mesh does not demonstrate asymptotic order. Native steady-equation checks, Ghia errors and recorded initial failed attempts remain available with the data.

Computational cost. Fine grids and higher Reynolds numbers may require substantially longer runtimes and more memory. Students should begin with the supplied coarse case and estimate cost before selecting 64 or 128 cells on a laptop or Colab runtime.

21 A measured nonlinear coupling effect in the cavity

Method	Δt	Outer	Inner	Nonlinear momentum defect	Continuity defect
PISO	0.01	1	1	1.61×10^{0}	5.90×10^{-14}
PISO	0.01	1	2	1.78×10^{-1}	4.20×10^{-14}
PISO	0.01	1	4	1.78×10^{-1}	4.26×10^{-14}
PIMPLE	0.01	3	2	7.26×10^{-6}	5.23×10^{-14}
PIMPLE	0.01	6	2	1.14×10^{-12}	6.84×10^{-14}
PISO	0.05	1	1	5.13×10^{0}	7.27×10^{-14}
PISO	0.05	1	2	1.11×10^{0}	2.38×10^{-14}
PISO	0.05	1	4	1.12×10^{0}	2.96×10^{-14}
PIMPLE	0.05	3	2	3.43×10^{-3}	4.92×10^{-14}
PIMPLE	0.05	6	2	3.75×10^{-7}	4.93×10^{-14}
PISO	0.2	1	1	9.34×10^{0}	1.37×10^{-13}
PISO	0.2	1	2	2.39×10^{0}	8.88×10^{-15}
PISO	0.2	1	4	2.47×10^{0}	1.14×10^{-14}
PIMPLE	0.2	3	2	4.92×10^{-2}	9.77×10^{-15}
PIMPLE	0.2	6	2	1.46×10^{-4}	3.38×10^{-14}

Fifteen configurations start from rest at $Re = 100$ on 32 cells per side. Each takes one backward-Euler step. At each fixed Δt, the study varies inner pressure corrections and outer coefficient updates.

The nonlinear step diagnostic rebuilds convection from the final velocity, while the frozen-predictor diagnostic uses the coefficients assembled for the predictor. Extra PISO corrections can reduce pressure–velocity inconsistency without eliminating the error from frozen nonlinear coefficients. PIMPLE also updates those coefficients, with additional momentum assembly and solves.

All retained cases in this one-step study satisfy a cell-divergence defect below 10^{-9}. The remaining nonlinear momentum defects distinguish the correction strategies. Their differences demonstrate algebraic coupling behavior for this setup, not the accuracy of a transient trajectory.

Each panel has its own logarithmic vertical range. For a temporal-accuracy study, reduce Δt and compare at matched physical times. More correctors and a stable larger time step do not establish an accurate startup history or a universal wall-time advantage.

22 Notebook controls and student analysis

The notebook presents the same derivations and runnable algebraic examples, followed by retained CFD results. Fresh CFD runs are opt-in so the default lesson can be inspected without repeating every expensive calculation.

Control	Purpose
`RUN_NEW`	Fresh baseline calculation at $Re = 100$
`RUN_HIGHER_RE`	Fresh experimental higher-Re calculation
`HIGHER_METHOD`	One method or all four formulations
`HIGHER_CELLS`	Number of cells per side / matched FD spacing
`RUN_STREAMFUNCTION`	Independent FD refinement experiment

Conservation. Derive the four-face cell pressure equation and verify the two-cell example. Explain why a zero global flux does not guarantee zero cell imbalance. Add a constant pressure offset and check that velocities do not change.

PISO. Reproduce both corrections of the frozen-matrix example. Identify which quantities change between correctors. Explain why zero divergence after the first correction does not imply zero momentum defect.

PIMPLE. Reproduce the three outer passes of the nonlinear example. Evaluate the nonlinear defect using coefficients rebuilt from the final velocity. Explain why the old-time source must stay fixed. Compare one outer pass with PISO.

Cavity comparison. Start at $Re = 100$ and matching mesh spacing. Record final cell divergence, unrelaxed momentum defect, both Ghia profile errors, runtime, and solve counts. Change one coupling setting at a time; keep spatial equations and final accuracy targets fixed.

Higher Reynolds number. At $Re = 400$ and 1000, separate spatial refinement from coupling changes. Retain failed configurations. State the cost before proposing a fine-grid run. Native FD and FV residual magnitudes must not be interpreted as the same error.

Transient extension. Specify a final physical time, run a sequence of halved time steps, and compare common field or phase errors. Keep the time scheme fixed when comparing coupling algorithms. The current steady SIMPLE iteration is not a transient trajectory.

23 References

The textbook treatments develop finite-volume conservation, staggered pressure–velocity coupling and pressure correction. The paper and official algorithm documentation provide the PISO origin and practical inner/outer-loop distinctions.

[1] S. V. Patankar (1980). *Numerical Heat Transfer and Fluid Flow*. Hemisphere/Taylor & Francis. Publisher. Foundational pressure correction, staggered grids and SIMPLE.

[2] H. K. Versteeg and W. Malalasekera (2007). *An Introduction to Computational Fluid Dynamics: The Finite Volume Method*, 2nd ed. Pearson. Publisher. Finite-volume transport and pressure–velocity coupling.

[3] J. H. Ferziger, M. Perić and R. L. Street (2020). *Computational Methods for Fluid Dynamics*, 4th ed. Springer. doi:10.1007/978-3-319-99693-6. Incompressible-flow solution methods and numerical errors.

[4] F. Moukalled, L. Mangani and M. Darwish (2016). *The Finite Volume Method in Computational Fluid Dynamics*. Chapter 15, "Fluid Flow Computation: Incompressible Flows," pp. 561–654. Chapter DOI. Staggered/collocated arrangements and SIMPLE/PISO derivations.

[5] R. I. Issa (1986). "Solution of the implicitly discretised fluid flow equations by operator-splitting." *Journal of Computational Physics* 62(1), 40–65. doi:10.1016/0021-9991(86)90099-9. Original PISO formulation.

[6] CFD Direct, *OpenFOAM User Guide v13*, sections 4.6.7–4.6.8. Solution and algorithm control. Pressure reference, correctors and outer correctors. The teaching solver's fixed-loop MAC implementation is specified in the preceding equations; software variants need not have identical defaults.

[7] OpenCFD, *Pressure–velocity coupling*. Official equation decomposition. Momentum diagonal and neighbor-source representation.

[8] U. Ghia, K. N. Ghia and C. T. Shin (1982). "High-Re solutions for incompressible flow using the Navier–Stokes equations and a multigrid method." *JCP* 48, 387–411. doi:10.1016/0021-9991(82)90058-4. Velocity benchmark samples.

The examples use a staggered grid. Collocated-grid interpolation and Rhie–Chow stabilization are separate topics and are not implemented by this solver.